CSC148 Introduction to Computer Science

Fall 2024 Midterm, Part A: Multiple Choice. Circle one answer per question.

1. What does `f(4)` return?

```
def f(n):
    if n <= 1:
        return 1
    return n * f(n - 2)
```

A. 24

B. 8

C. 4

D. 6

2. What is the worst-case running time of binary search on a sorted list of n items?

A. O(1)

B. O(log n)

C. O(n)

D. O(n log n)

3. Given only a reference to the head node, what is the cost of inserting a new node at the front of a singly linked list of n nodes?

A. O(1)

B. O(log n)

C. O(n)

D. $O(n^2)$

4. Starting from an empty stack, we run push(1), push(2), pop(), push(3), push(4), pop(). Which value is now on top?

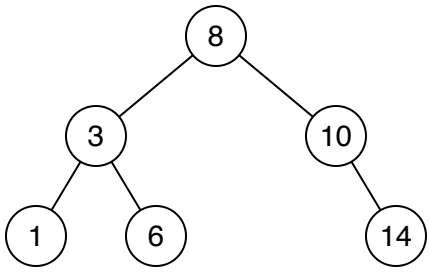
A. 1

B. 2

C. 3

D. 4

5. The binary tree below is traversed in preorder. What is the third value visited?



A. 3

B. 1

C. 6

D. 10

Answer Key, Part A

For instructor use. Question 4 was dropped from the key after a printing error.

Question	Answer
1	B
2	B
3	A
5	B